

**Studies on Management of Dry Root Rot of Chickpea
caused by *Rhizoctonia bataticola* (Taub.) Butler**

THESIS

Submitted to the

Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur

**In partial fulfilment of the requirement for
the Degree of**

MASTER OF SCIENCE

In

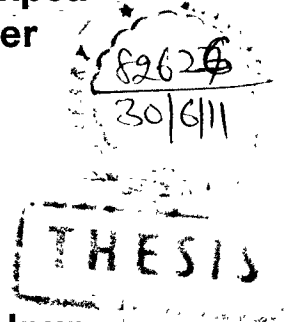
**AGRICULTURE
(PLANT PATHOLOGY)**

By

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Jabalpur (M.P.)**

2009



CERTIFICATE - I

This is to certify that the thesis entitled "**Studies on management of dry root rot of chickpea caused by *Rhizoctonia bataticola* (Taub.) butler**" Submitted in partial fulfillment of the requirement for the degree of "**MASTER OF SCIENCE**" in **PLANT PATHOLOGY** of Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur is a record of the bonafide research work carried out by **SHSHANK RANJAN** under my guidance and supervision. The subject of the thesis has been approved by the student's Advisory Committee and the Director of Instructions.

No part of the thesis has been submitted for any other degree or diploma (Certificate awarded etc.) or has been published/published part has been fully acknowledged. All the assistance and help received during the course of investigation has been acknowledged by him.

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Place: Jabalpur



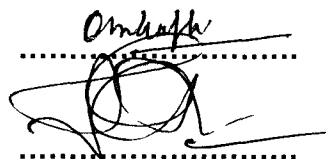
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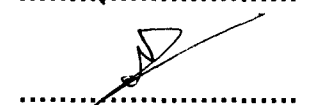
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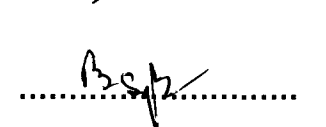
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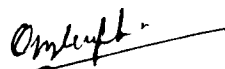


CERTIFICATE – II

This is to certify that the thesis entitled "**Studies on management of dry root rot of chickpea caused by *Rhizoctonia bataticola* (Taub.) butler**" Submitted by **Mr. SHSHANK RANJAN** to the Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur in partial fulfillment of the requirement for the degree of "**MASTER OF SCIENCE IN AGRICUTURE**" in **PLANT PATHOLOGY** in the Department of Plant Pathology has been ,after evaluation, approved by the External Examiner and by the student's Advisory Committee after an oral Examination on the same.

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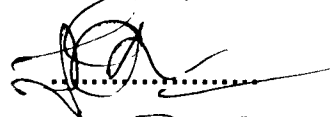
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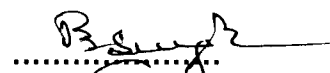
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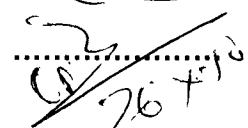
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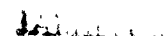


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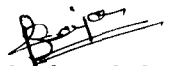
I own an everlasting depth of gratitude to my ideal and worshipful Grandfather **Shri Deeyali Patel**, my Grandmother (late) **Smt. Pyaree Patel**, my father **Shri Ayodhya Prasad Patel**, my mother **Smt. Mankunwar Patel** no words can explain their deep love for me in the long course of my education. They always gave me a helping hand and constant encouragement with a smile of love and affection. They waited patiently for the completion my Post Graduation.

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Place : Jabalpur

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(Shshank Ranjan)

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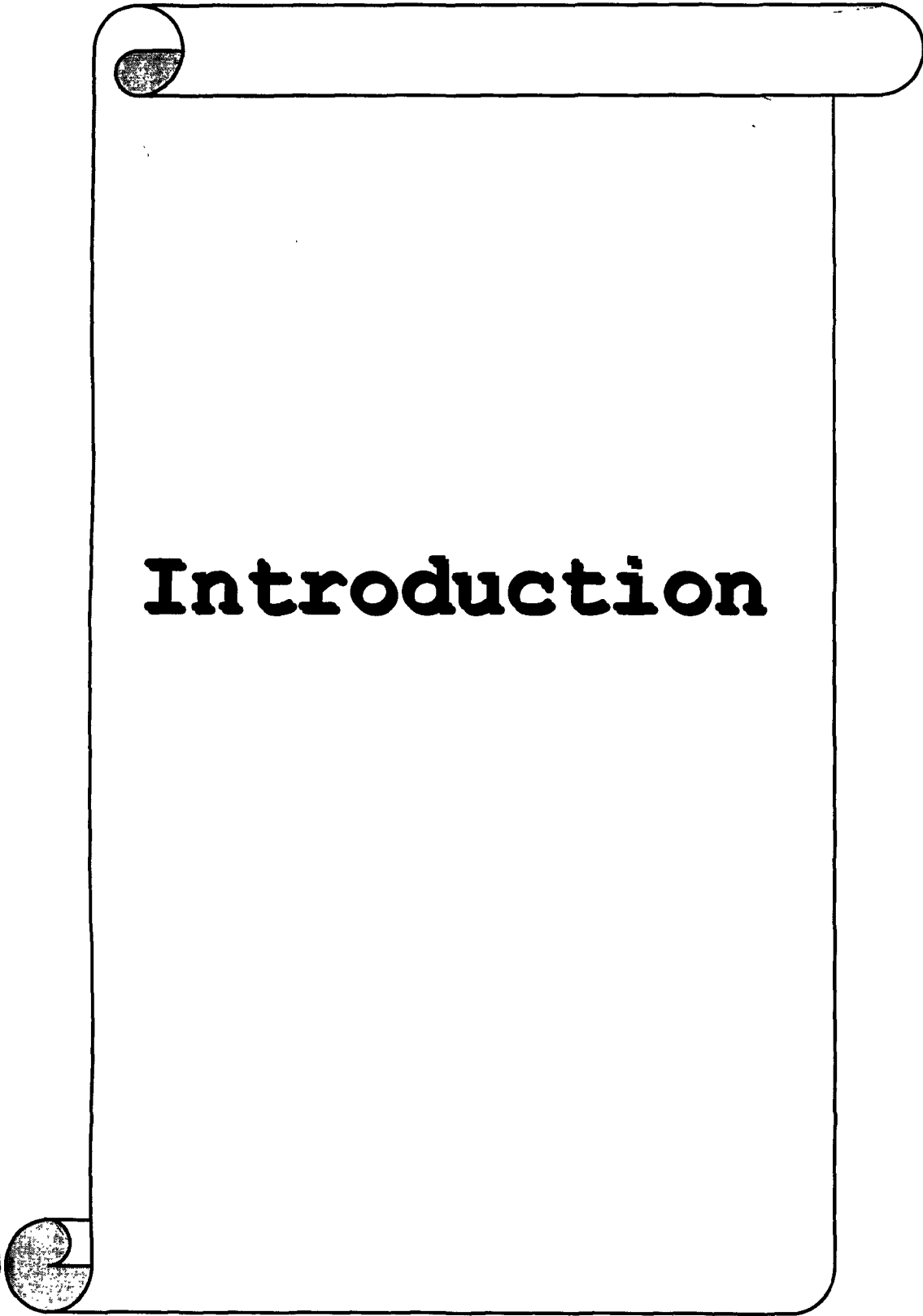
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ABBREVIATION

Cm	=	Centimeter
DRR	=	Dry root rot
En	=	Entries
Mm	=	millimeter
g	=	Gram
lb	=	Pound
psi	=	Per square inch
PDA	=	Potato dextrose agar medium
ml	=	milliliter
Lit	=	liter
No.	=	number
i.e.	=	that is
Sp.	=	species
°C	=	Degree celsius
etc	=	et cetera
<i>et al.</i>	=	Co-worker
eg	=	as for example
BOD incubator	=	Biological oxygen demand incubator
WP	=	Watable powder
ha	=	hectare
%	=	Per cent
@	=	at the rate of
DAS	=	days after sowing
Viz.	=	namely

A hand-drawn illustration of a scroll. The scroll is unrolled, showing a large rectangular area. The top edge is rolled up, and the bottom edge is also rolled up. The word "Introduction" is written in a bold, black, serif font in the center of the scroll. The scroll has a light beige or cream color with a slightly textured appearance. The edges of the scroll are outlined in black, and the rolled-up parts are shaded with a stippled effect.

Introduction

INTRODUCTION

In Indian Agriculture pulses play an important role not only from economical point of view but also due to their nutritional value. Chickpea is valued for their nutritive seeds with high protein content, 25.3 - 28.9 per cent after dehulling (Hulse, 1991) carbohydrate - 61.5 per cent, fat - 4.5 per cent and vitamin – 2.4 mg per 100 g. Chickpea is grown in tropical, subtropical and temperate regions. The total cultivated area of chickpea in the world is about 20.05 m.hac. and annual world production is 11.14 m. tonnes. The average yield / hac. is estimated to be around 556 kg/ha. (Ramana Rao, 2005).

The major chickpea growing countries are India, Pakistan and Turkey in Asia, Ethiopia in Africa, California and Washington State in the U.S.A., Mexico and Australia (F.A.O. 1994). In India chickpea is cultivated in 7.63 m.hac. with an annual production 5.97 m.tonnes and the average yield is estimated 782 kg / hac. (Agri. Statistics at a glance, 2006-07).

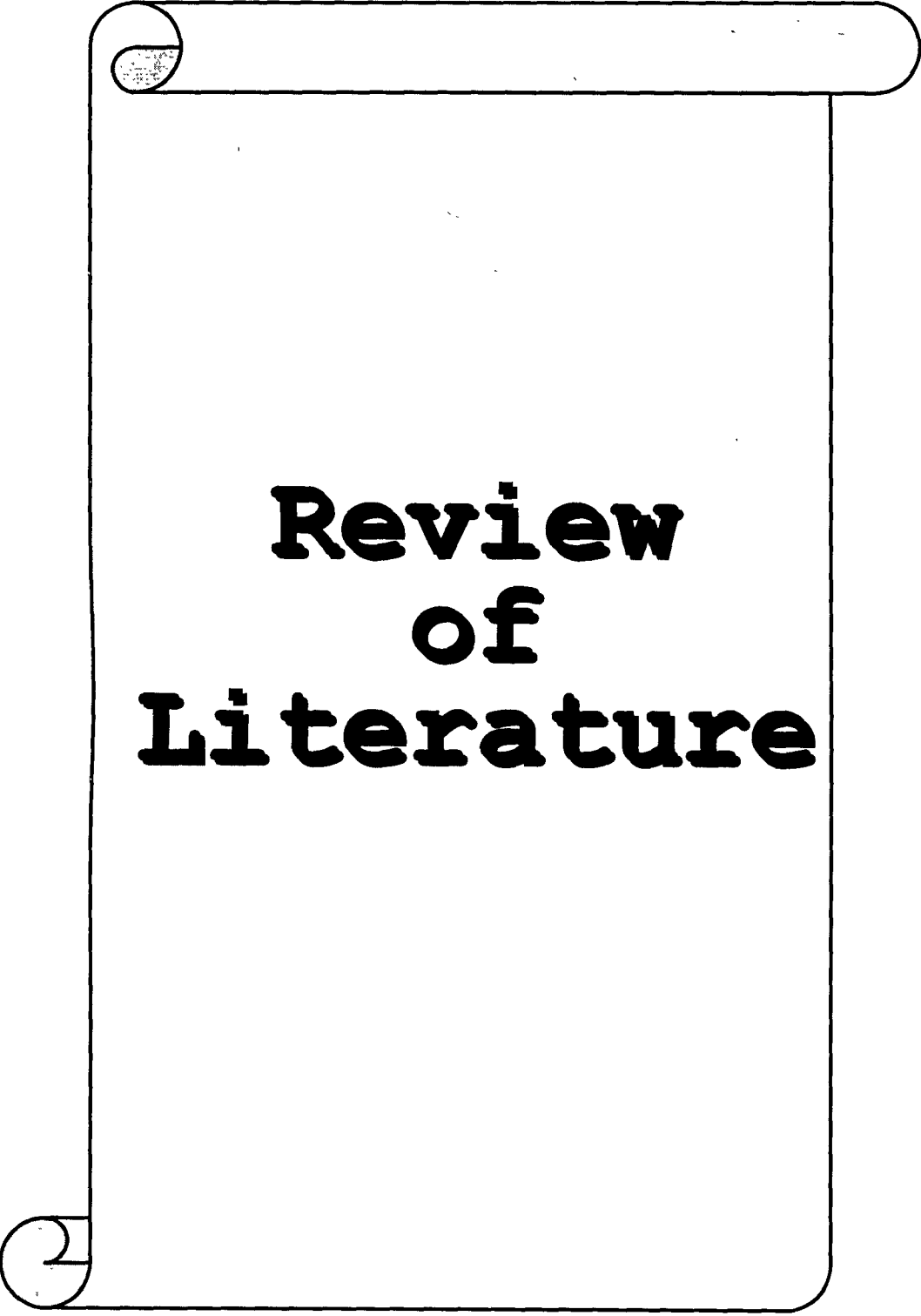
In India the major chickpea growing states are Madhya Pradesh, Rajasthan, Maharastra and Gujarat. Madhya Pradesh contributes 46.2 per cent of the total production of chickpea in the country. In Madhya Pradesh it ranks 1st in both area (2.56 m.hac.) and production (2.37 m.tonnes) and the average productivity is estimated 926 kg / hac. (Agri. Statistics at a glance, 2005-06).

There are several chickpea varieties which are divided mainly in three categories viz, desi, gulabi, and kabuli type which are grown as sole / mixed crop with wheat, mustard, lentil, etc. Several seed, soil and air borne diseases are responsible for lowering its yield. Chickpea is infected by 67 fungal pathogens of which *Fusarium oxysporium f.sp. ciceri* (Wilt), *Rhizoctonia*

bataicola (Dry root rot) and *Sclerotium rolfsii* (Collar rot) are an important diseases in Madhya Pradesh.

The first report of occurrence of root rot in chickpea alongwith wilt was made by Padwick (1948). The species *Rhizoctonia bataicola* is responsible for dry root rot which is favoured by high temperature (>30 c) and mostly occurs when the crop is in flowering/pod formation stage. The affected plant shows straw colour apperance and roots become black and brittle,which can broken easily.

The review of literature of this disease reveals that workers have reported disease management through chemical, biological and host resistance but little work has been done to attain a strategy towards integrated disease management using commercially available biocontrol agent .Hence, investigations were undertaken with an objective to find out an effective strategy by integrating the most effective components i.e. Bioagents (*Trichoderma viride*) fungicides and host resistance for the management of dry root rot of chickpea.



Review of Literature

REVIEW OF LITERATURE

The first mention of a plant disease, which may be referred with, certainly to *Rhizoctonia* as a causal agent is an important paper by Du Hamel (1728) read before the Paris Academy. In this paper he gave a careful description of a fungus disease of saffron (*Crocus sativus*) occurring in France. He named the fungus *Tuberorides*.

De Candolle (1815) discovered a similar disease on alfalfa. In his taxonomic description he was first person who gave the name for the fungus. Later, after careful study of the characteristics and parasitism of the species, he established (1815) the genus *Rhizoctonia*.

In 1924, Small described a *Rhizoctonia*, which develop black sclerotia of variable size and shape in the diseased roots, many woody plants from Uganda and he named it *Rhizoctonia lamrllifera*. Butler (1925), how ever, considered that the fungus should be regarded as *Rhizoctonia* and made the combination *Rhizoctonia bataticola* (Taub.) Butler to a *sclerotium* causing a root rot of cotton, french beans and cowpeas.

Taxonomy

The most important taxonomic work on the genus *Rhizoctonia* in the 19th century is that of Tulasne brothers (1862). They published the first extensive morphological study on the genus *Rhizoctonia*. They reduced *Rhizoctonia crocorum* DC. and *Rhizoctonia medicaginis* DC. to a single species to which they applied a new name *Rhizoctonia vivolaceae* Tul. This reduction to a single form was made after a most careful morphological study of the fungus at all stages.

Among the taxonomic and pathological studies on *Rhizoctonia* in India, the most important is that of Shaw. Before 1912, he had studied two species of *Rhizoctonia*, one of which was characterized by its formation of small black sclerotia. The systematic position of *Rhizoctonia bataticola* as follows (Alexopoulos and Mims 1983).

Kingdom	-	Mycetae
Division	-	Amastigomycota
Sub-division	-	Deuteromycotina
Class	-	Deuteromycetes
Sub-class	-	Coelomonycetidae
Order	-	Spharopsidales
Family	-	Spharopsidaceae
Genus	-	<i>Macrophomina</i>
Species	-	<i>Phaseolina</i>

Pathogen

The species description of *R. bataticola* as follows (Subramanian, 1971)). "The basic characteristics of the species are the formation of sclerotia in the host tissue as well as in culture. On the infected plants sclerotia seen in the pith region and outer surface of roots just below the bark region. They are brown or jet black, minute, smooth externally, hard, variable in shape: globose, oval, oblong, elliptical, curve or even forked and varying in size, but of uniform texture, more or less loosely packed."

The cell of hyphae are barrel shaped, branching more or less at right angles and light to dark brown in colour, composed of free thick-

walled cells. The species *Rhizoctonia* is distinguished by the light brown-to-brown hyphae with somewhat loose/spongy sclerotia in the form of bulbils. According to Mehrotra and Aneja (1990) both the species *R. bataticola* and *R. solani* have the variable capacity to produce enzymes.

Jharia (1970) also found that *Rhizoctonia bataticola* is associated with seeds of urid both internally and externally. Thus it seems that *Rhizoctonia bataticola* acts as both soil and seed borne pathogen from Madhya Pradesh.

Reddy *et al.* (1991) stated that the countries like Mandalay, Sayaing and Bago, central and lower Myanmar the major disease in chickpea was dry root rot caused by *R. bataticola*. Aung *et al.* (1992) observed that dry root rot caused by *R. bataticola* was major disease.

Symptoms

Rhizoctonia bataticola is a soil borne fungus/pathogen. Several workers have reported various types of symptoms produced on plants with varying intensity of the disease in different crops and different regions.

According to Mehrotra and Aneja (1990) the first symptom of the disease is yellowing of the leaves within a day or two such leaves drop and in the course of the next two or three days whole plant dies off. The leaves and stem of the affected plants are usually straw coloured. The taproot is dark and quite brittle in dry soil.

Singh *et al.* (1990) observed the roots inoculated with *R. bataticola* [*Macrophomina phaseolina*] and reported disintegration of cortical tissues while mycelium and sclerotial bodies plugging the xylem were seen in longitudinal sections.

Occurrence and losses

The first report of the occurrence of root rot in chickpea along with wilt was made by Padwick (1948). The incidence of root rot ranging from 3.58 to 20.63 per cent in 30 villages of northern Madhya Pradesh was reported by Gupta *et al.* (1983). This is one of the most common diseases in South East Asian countries and limits chickpea production (ICRICAT Annual Report 1986: FAO, 1990).

Nema and Khare (1973) reported that chickpea cultivation in Madhya Pradesh was greatly threatened by soil borne diseases generally appearing at two phases of growth, early seedling stage and flowering stage, wilting was due to *Fusarium oxysporum* f. sp. *ciceri*, dry root rot caused by *Rhizoctonia bataticola* and *Sclerotium rolfsii* resulted in collar rot.

Tripathi and Sharma (1983) studied the incidence of chickpea dry root rot (*Rhizoctonia bataticola*) in Southern Haryana and reported that the incidence was high from late October to mid November which was described in December and January and then again increased in the months of February and March. They also noted that seed sown at a depth of 15 cm developed a less root rot than that sown at 7.5 cm.

Singh *et al.* (2002) conducted a survey in Vindhyan plateau zone of Madhya Pradesh, India to evaluate the prevalence of wilt and dry root rot disease in chickpea. They reported that *Fusarium oxysporum* was most prevalent (56 per cent) in wilted plants followed by *Rhizoctonia bataticola* (17 per cent) and *Trichoderma* sp. (10 per cent). The incidence of *Rhizoctonia bataticola* also caused withering and drying of plants, which ranged from 8 to 20 per cent.

Pathogenicity

Dermiro *et al.* (1994) observed that dry root rot disease of gram is also becoming a serious disease affecting productivity of this crop. They isolated *Rhizoctonia bataticola* from plants and seeds of chickpea cultivars Azizive 94 and tested pathogenicity.

Byadgi and Hedge (1985). Studied the variation in size and shape of pycnidial and pycnidiospores of *M. Phaseolina* on chickpea based on morphological cultural characters and pathogenic behaviour.

Evaluation of fungicide and biological agent (*Trichoderma viride*) on *Rhizoctonia bataticola* under *in vitro* and *in vivo* condition

Fungicides

Verma. (1976) found seed treatment with Bavistin (Carboxin) @ of 0.25 per cent to protect seedling in field for 30 days or more from root rot of chickpea.

Goyal and Mehrotra .(1981) tested 9 fungicides against gram isolates of *Rhizoctonia bataticola*(*M. phaseolina*) *in vitro* and found Benlate, Bavistin Thiophanat-methyl to be promising.

Taneja and Grove .(1982) tested four fungicides against *R. bataticola in vitro* and found Benomyl and Bavistin to be best.

Taya et al .(1990) reported that in screen house test the best control of Dry root rot caused by *Rhizoctonia bataticola* (*Macrophemina phaseolina*) of the 8 fungicides tested was given by Carbendazim alone or combination with Thiram as seed treatment. Per sowing soil dranch and seed treatment +drenching after sowing of chickpea.

Mathieson and Rush.(1991) observed that seed treatment with triadimenol had significantly greater emergence and dry matter production in chickpea in root rot infested field.

Singh *et al* .(1998) reported that Carbendazim and Carboxin (0.2%) used as seed treatment, soil drenching and seed treatment+soil drenching significantly reduce the incidence of Dry root rot of chickpea.

Sindhan *et al* .(1998) found that Bavistin and Vitavax(0.2%) used as seed treatment, soil drenching and seed treatment+ soil drenching significantly reduce the incidence of Dry root rot of chickpea with corresponding increase in Phenolic compound and Carbohydrate contents of the plant.

Lambhate *et al* .(2002) evaluated the efficacy of Bavistin (Carbendazim), Blitox-50 (Copper oxychloride), Canton, Ridomil (metalaxyl), Thiram, and Topsin-m(Thiophanat-methyal) *in vitro* at 0.1, 0.2, and 0.3 per cent against *M. phaseolina*.

Prajapati *et al* .(2002) conducted studies to determine the efficacy of 11 fungicides, 8 nontargant pesticides and 2 biological control agents against the causative agent of Dry root rot in chickpea. Carbendazim, Carbendazim+Thiram, Carboxin and Topsin-m (Thiophanat-methyal) completely inhibited the growth of *R. bataticola*.

Kaushal and Sood (2008) reported that seed treatment with thiram and bavistin@ 2 g/kg seed gave good control of root rot of chickpea under field condition.

Bioagent

Deshmukh and Raut (1992) conducted *in vitro* test and found that *T.harazianum* and *T. viride* overgrow on colonies of *R. bataticola* .

Saxena and Saxena (1995) reported the use of fungal antagonists like *T. viride* and *T.harazianum* was found effective against root rot of chickpea caused by *M. phaseolina*.

Selvarajan and Jeyarajan (1996) reported that *T. viride*, *T.harazianum* and *B. subtilis* formed inhibition zones against chickpea root rot pathogens (*F. solani* and *M. phaseolina*).

Singh *et al* (1998) reported *Trichoderma viride* and *T.harazianum* used as seed treatments and soil inoculum proved effective in reducing the incidence of dry root rot (*Rhizoctonia bataticola*) and increasing the phenolic and carbohydrate contents of chickpea.

Kumar *et al.* (2000) reported that seed treatment with *Trichoderma viride* and *Chaetominum globosum* was used to control seed brone fungal pathogen. *T. viride* completely eliminated seed brone infection of *Alternaria alternata*, *Phyllostica cajceni*, *Rhizoctonia bataticola* and *R. solani*.

Sindhan *et al.*(2002) studied the efficacy of *Trichoderma viride*, *T.harazianum* , *Aspergillus flavus*, *A. niger*, *A. ochraceus*, *Penicillium citrinum*, *Bacillus subtilis* and *Pseudomonas fluorsacens* against *Rhizoctonia bataticola* under *in vitro* and green house condition. *In vitro*, all antagonists inhibited mycelial growth and sclerotia production with *P. fluoresens* being the most effective. However, Carbendazim was more effective than the bacterial antagonists, inhibiting mycelial growth by 94.2% and sclerotia production by 100 Per cent.

Suryawanshi and Mandhare.(2009) reported that wilt caused by *Fusarium oxysporum f. sp. ciceri*, Collar rot by *Sclerotium rolfsii* and dry root rot by *Rhizoctonia bataticola* are the important soil brone pathogen that infect the chickpea crop. Experiment on efficacy of botanicals and

Trichoderma sp. against these soil borne diseases was undertaken in laboratory. *Trichoderma lignorum* and *T. virens* inhibited the growth of *S. rolfsii* by 68 and 67%, respectively while the growth of *R. bataticola* was inhibited to the extent of 76% by *T. viride* and *T. harzianum*.

Host resistance

Ananda Rao and Haware. (1987) identified several lines of chickpea resistant to dry root rot at ICRISAT. Some of these lines have the resistance to wilt and dry root rot. The inheritance of resistance to dry root rot of chickpea seems to be monogenic with resistance dominant over susceptibility.

Nene *et al.* (1989) identified broad based stable resistant lines to wilt and root rots on the basis of multilocation evaluation, test (ICC 11550, ICC 9032, ICC 9023, ICC 2862 and ICC 11551).

Reddy *et al.* (1990) reported two chickpea varieties *i.e.* Avrodhi and ICC 98 resistant to wilt and root rots from ICRISAT during 1985-86 to 1988-89.

Bhatt (1993) observed that BGM 416, BG 416, ICC 1376 and IC 11314 were resistant to root rot. Resistant cultivars had greater number of total roots during early growth phases.

Gupta (1997) identified genotypes ICCV 89402, ICCV 90201, ICCV 90254, BG 209, H 88-2, GF 88426 and RSG 180 having dual resistance against wilt and dry root rot.

Yadava *et al.* (2000) Eighty-one chickpea genotypes were evaluated for their reaction to Wilt (*Fusarium oxysporum f.sp. ciceris*), root rots (*Rhizoctonia bataticola*), *Sclerotium rolfsii* and *F. solani* in the field. There were 10 genotypes resistant to these diseases, with JG 22,

JG 33, KPG 316 and SAKI 93130 showing a stable resistant against dry root rot.

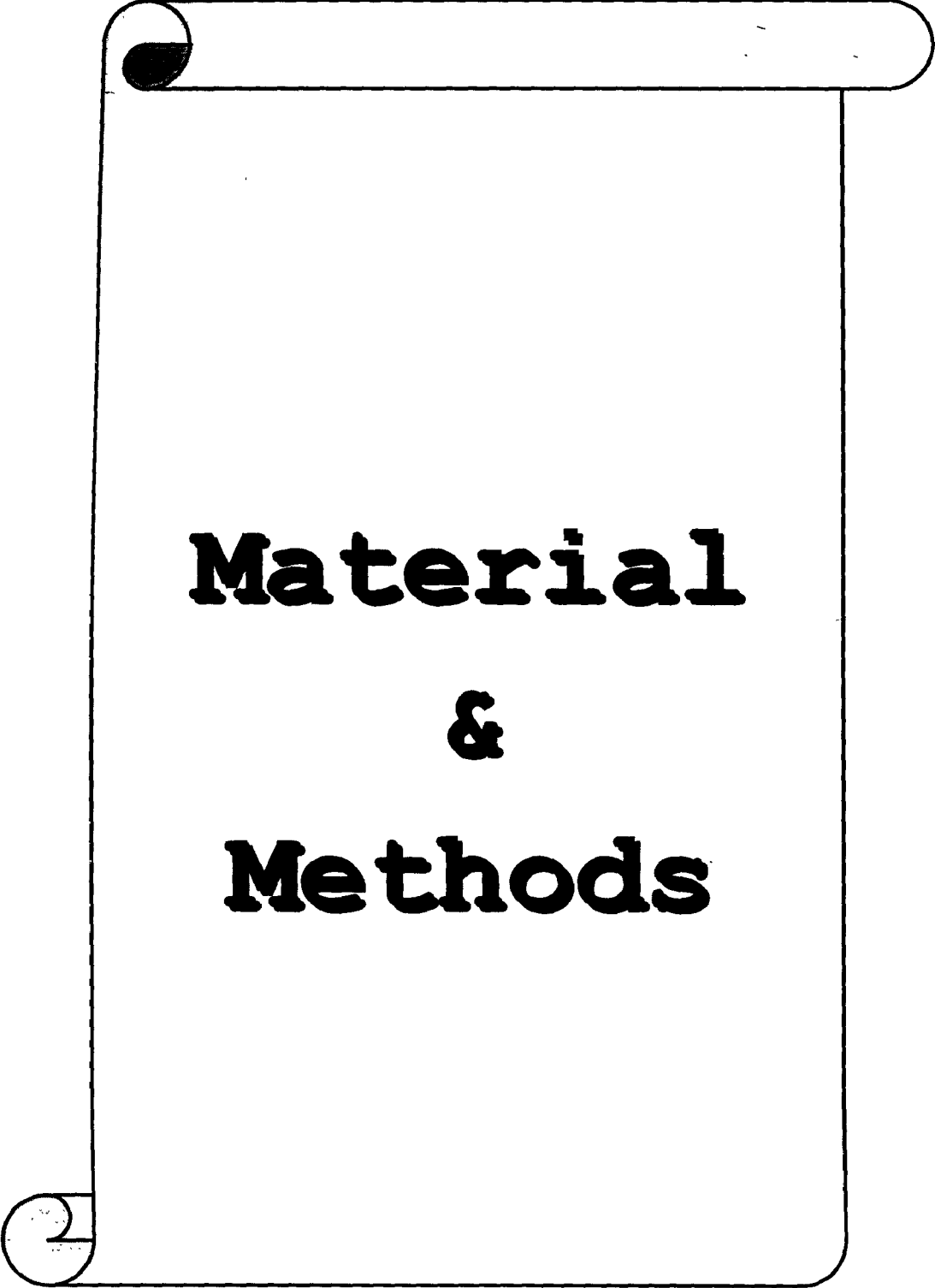
Gangwar *et al.* (2002) screened 35 chickpea cultivars for resistance to dry root rot caused by *Rhizoctonia bataticola* in a field experiment. All the cultivars exhibited disease symptoms, only 10 cultivars (ICC 2644, 10384, 10630, 112244, 11332, ICCL 81002, 810810, ICC 12263, 12441 and ICCV 90254) were resistant whereas 5 genotypes (GCP 9504, Phule G 96020, 96105, 96313 and GL 91059) were moderately resistant to dry root rot.

Prajapati *et al.* (2003) while working on resistant sources of chickpea against dry root rot (*Rhizoctonia bataticola*) evaluated forty-five chickpea cultivars for resistance to dry root rot. Among these, H 355, JG 315, Pusa 212, G 543, ICC 32, BG 244, Phule G 5, IICC numbers 2644, 10384, 11224, 10630 and 11332, ICCV 10, ICC numbers 12441 and 12263, 81010 and 81002, Avrodhi, ILC 702 and ILC 200 cultivars were rated as resistant. 5 cultivars were moderately resistant and 9 were tolerant.

Panday *et al.* (2004) evaluated twenty-nine chickpea germplasm accessions, 10 cultivars and 8 advanced breeding lines for resistance to dry root rot, caused by *Rhizoctonia bataticola* under *in vitro* conditions. They observed that one germplasm accession (ICC 14395), a cultivar (ICCV₂) and advanced breeding line were resistant to dry root rot. Of the remaining lines 22 were moderately resistant, 19 susceptible and 2 highly susceptible line (BG 212 and ICC 12267) used as the control.

Gupta and Babbar (2006) conducted a field experiment to search the multiple as well as individual disease resistant sources against

three soil borne diseases in chickpea. The results revealed that the pathogens *Sclerotium rolfsii* and *R. bataticola* caused severe mortality ranging from 1-74 and 1-100 percent at seedling and later stages of crop growth. However, the genotypes, which performed well, exhibited <10 per cent mortality against these diseases have been listed.



Material & Methods

MATERIAL AND METHODS

Present investigations were undertaken to study the “Management of dry root rot of chickpea caused by *Rhizoctonia bataticola* (Taub.) Butler.” The material used and methods followed are described below:

(A) MATERIAL

1. General

Borosil make glassware's were used throughout the investigation. These were cleaned with cleaning solution followed by thorough washings and rinsing with water. Glassware's were sterilized in hot air oven at 180⁰ C for two hours.

2. Test pathogen

Rhizoctonia bataticola the causal organism of chickpea dry root rot.

3. Culture Media

The following media were used during the course of investigation.

(a) Potato dextrose agar (P.D.A)

Peeled potato	-	200 g
Dextrose	-	20.0 g
Agar	-	20.0 g
Distilled water	-	1000 ml

(b) Potato dextrose broth (P.D.B.)

Peeled potato	-	200 g
Dextrose	-	20.0 g
Distilled water	-	1000 ml

For potato extraction 200g peeled and sliced potato was taken and boiled in 500 ml water. Then it was strained through muslin cloth. Agar-agar was washed with water and taken in 500 ml of water then heated until it melted fully. Melted agar-agar was poured in potato extract and 20 g dextrose was added. The solution was finally make-up to one liter. The media was poured in conical flask and sterilized in an autoclave at 15 lbs., p.s.i / at 121.6⁰C for 20 minutes.

In Potato dextrose broth similar procedure as described above was employed only agar-agar was not added in potato dextrose broth.

(B) METHODS

Collection of diseased plant / root samples

The plant samples showing the symptom of dry root rot were collected from the research experiments conducted at Seed Breeding Farm under All India Coordinated Research Project on chickpea.

Isolation of *Rhizoctonia bataticola* from plant samples

The roots of diseased chickpea plant showing root rot symptoms were washed thoroughly with tap water, small pieces from infected roots were cut with the help of sterilized blade. These pieces were surface sterilized with 1:1000 mercuric chloride (HgCl₂) solution for one minute followed by three changes in sterilized distilled water to remove traces of HgCl₂. The pieces were then transferred aseptically to petriplates containing PDA. Inoculated petriplates were incubated at 28 ± 1⁰C for 3-5 days and examined at frequent intervals to see the growth of the fungus developing from different pieces.

Purification and identification

The cultures of *R. bataticola* was purified by subculturing the single hyphal tip method and maintained by mass transfer on potato dextrose agar medium at room temperature. After purification, the pathogen was identified by observing the colony against light with the naked eyes and later confirmed with the help of microscope.

Pathogenicity test

To conduct the pathogenicity test Blotter paper technique as described by Nene *et al.* (1981) was employed.

Evaluation of fungicides on growth and sclerotia formation of *Rhizoctonia bataticola*

Seed dressing fungicides viz., Vitavax (Carboxin), Bavistin (Carbendazim), and Insignia (Polyclostrabin) were tested against *Rhizoctonia bataticola* by poisoned food technique. Bavistin (Carbendazim) and Vitavax (Carboxin) were tested at 0.2 per cent concentration and Insignia fungicide was tested at 0.1, 0.15 and 0.2 per cent concentrations. The required quantity of fungicide was diluted in a test tube containing 10 ml of sterilized water and mixed thoroughly to make suspension homogenous. This fungicides suspension was poured in a flask containing 100 ml media.

Fifteen milliliter of above fungicides added medium was poured in sterilized petriplates in six replications and check was kept without fungicide. Five mm disc of seven days old culture of *R. bataticola* was placed after 24 hours. The plates were incubated at 30°C for 10 days and observations were recorded on growth and sclerotia formation of fungus. Slides were prepared from 10 days old culture of *R. bataticola*. Sclerotia

were counted/microscopic field from each treatment under high power (40 x) and average of 10 observation were calculated.

Evaluation of commercial products of *Trichoderma viride* by the dual culture technique

Three commercial products of *Trichoderma viride* (biocontrol agent) were procured from market for studying their efficacy against *R. bataticola*.

The details are as follows

S.No	Product	Biocontrol agent
1.	Neemoderma	<i>Trichoderma viride</i>
2.	Niprot	<i>Trichoderma viride</i>
3.	Trioseal	<i>Trichoderma viride</i>

Measurement of CFU (colony forming unit)

First of all, the products were examined for their viable count in terms of CFU. For this purpose 1 g of each product sample was taken and added into 10 ml of distilled water in a culture tube. Serial dilutions were made by taking 1 ml of suspension and added to another 9 ml of water in a culture tube which were expressed as 10^{-1} , 10^{-2} , 10^{-3} 10^{-4} .

Extreme dilutions (10^{-3} , 10^{-4}) were utilized for further testing. One ml of each dilution was poured on petriplates containing potato dextrose agar in three replications. These plates were incubated in inverted position at 30°C and observed for their colony growth. The number of colonies were counted on 3rd day, CFU was calculated.

Dual culture technique was used to evaluate the efficacy of commercial products of *Trichoderma viride*. Five mm disc of each culture was placed in petriplates opposite to disc of *R. bataticola* taken from 7 days old culture in three replication. Check was kept by placing disc of *R. bataticola* only at one side of petriplate. These plates were incubated at

were counted/microscopic field from each treatment under high power (40 x) and average of 10 observation were calculated.

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30°C for 10 days. The efficacy was determined on the basis of their ability of over growth, inhibition zone and lysis of mycelium of *R. bataticola* by scoring on a modified Bell's scale (Bell *et al.*, 1982).

Modified Bell's Scale

R ₁	=	100% overgrowth
R ₂	=	75% overgrowth
R ₃	=	55% overgrowth
R ₄	=	Block at point of contact
R ₅	=	Pathogen overgrow antagonist

Observation on mycelial growth and sclerotia formation were recorded on 3rd, 5th and 7th day.

Identification of host resistance

Blotter paper technique: (*in vitro*)

Blotter paper technique was employed (Nene *et al.*, 1981). For this purpose 5 mm disc of the culture was placed on PDA poured petriplates and incubated at 25°C for 4 days. Five mm disc was cut from the above culture and transferred to 250 ml flasks each containing 100 ml of P.D. broth. After 5 days of incubation at 25°C two mycelial mats were removed from the flask which were added to 100 ml sterilized distilled water in a beaker after its proper crushing for 1 minute in the blender.

Surface sterilized seeds of different chickpea genotypes were sown on plastic trays containing sterilized soil + sand (1:1). Five days old seedlings of each genotype were uprooted in such a way so that root system is not disturbed. The root system of these seedlings was properly washed in running water followed by rinsing in sterilized distilled water. The roots of all genotypes (test lines) were dipped in the inoculum kept in a beaker with an up and down movement for about 30 seconds and the excess inoculum was removed by touching the roots to the edge of beaker.

Twenty seedlings of the each test line were taken and 10 of these were kept separately on two blotter papers (size 45 cm x 25 cm with one fold). The blotter paper was moistened adequately and the seedlings were kept in such a way so that only cotyledons and roots are covered and the green tops of the seedlings remains outside the blotter paper. The seedlings of a susceptible check (BG 212) were also inoculated with each batch of test seedlings.

The folded blotter papers were kept one on top of the other, in heaps of ten in a tray. A susceptible check was kept in each heap. These trays were placed into the incubator at 35°C for 8 days. Artificial light was provided for 12 hrs and the blotter papers were moistened adequately on alternate day.

Rating scale

In order to find out the resistant genotype of chickpea, test lines were scored at the end of the incubation period by examining the seedlings for the extent of root damage and were scored for the disease on 1-9 point scale as mentioned below:

Rating scale used for disease scoring

Rating	Category	Symptoms of DRR	DRR%
1	Resistant	No infection on roots	0.0-10
2-3	Moderately resistant	Very few small lesions on roots	10.1-20
4-5	Tolerant	Lesions on roots clear but small, new roots free from infection	20.1-30
6-7	susceptible	Lesions on roots many, new roots generally free from lesions	30.1-40
8-9	Highly Susceptible	Roots infected and completely discolored	40.1 and above

Test lines showing ratings 1-3 are considered acceptable for the breeding programme, ratings 4-5 are acceptable only if lines with ratings 1-3 are not available, ratings 6-9 are not acceptable.

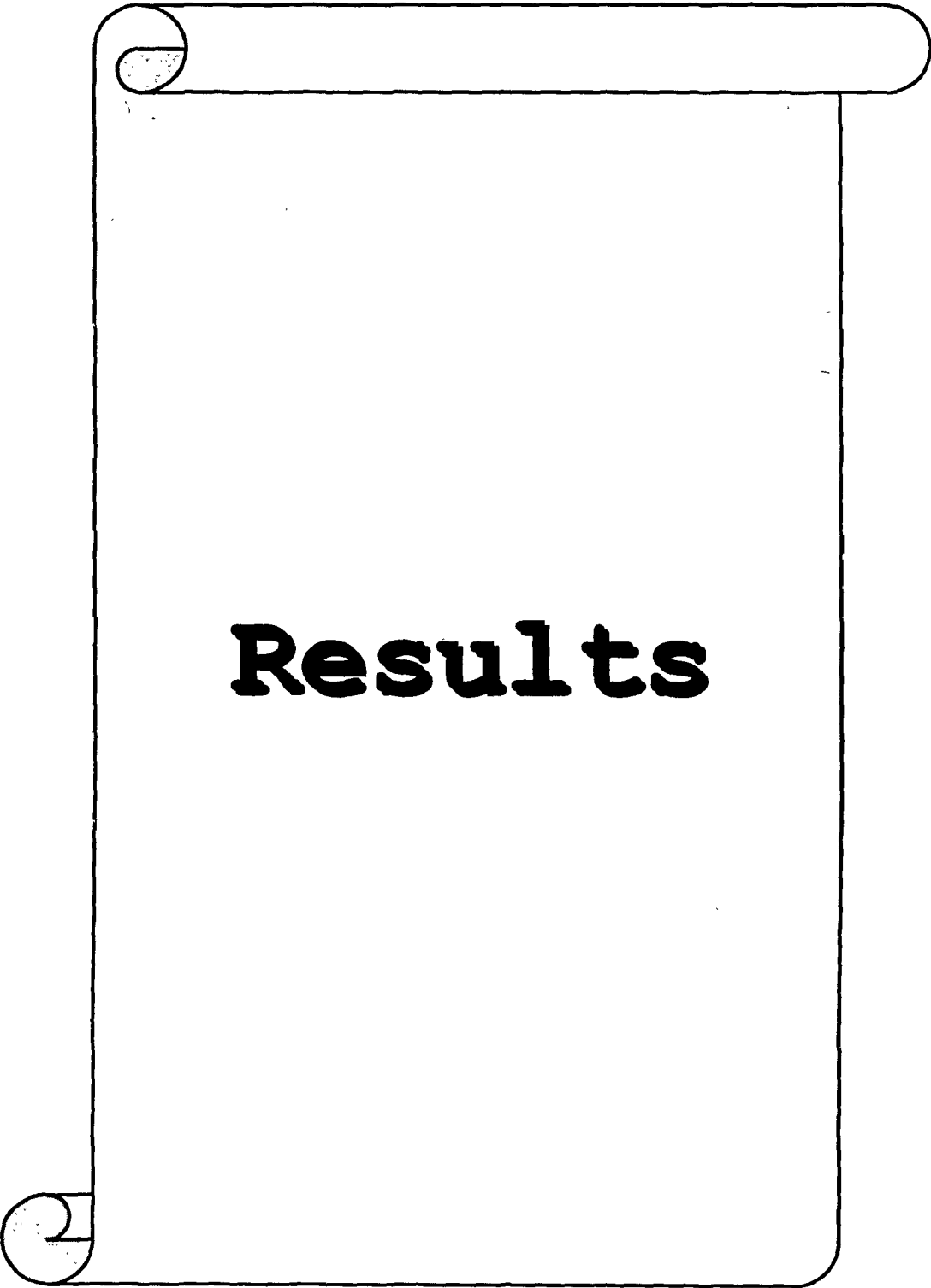
Disease sick field

Inorder to locate the sources of resistance, the chickpea lines were screened in disease sick field which was prepared by incorporating the culture of *R. bataticola* multiplied on chickpea straw.

The culture was mixed in the soil in a ratio of 1:9 and then added in each row before sowing. Forty seeds of each entry were sown in 4-meter long row at 30 cm apart. A susceptible check JG 62 after every two rows of test entry and one row of BG 212 in each trial was maintained for comparison. Observations on per cent emergence were recorded after 20 days of sowing and mortality due to dry root rot were recorded from flowering and podding stage.

Dry root rot percentage was calculated by the following formula.

$$\text{Percent of infected plants} = \frac{\text{Number of infected plants}}{\text{Total number of plants}} \times 100$$



Results

RESULTS

4.1 Collection of dry root rot infected plants

Dry root rot infected plants of chickpea were collected from disease sick field located at Seed Breeding Farm, Jawaharlal Nehru Krishi Vishwa Vidhyalay, Jabalpur, during February to March 2009.

4.2 Symptoms

Dry root rot infected chickpea plants were observed in the field at flowering to podding stage. The leaves and stems of affected plants showed straw-coloured appearance. The disease appeared in a severe form when ambient temperature was $>30^{\circ}\text{C}$. The pathogen forms reddish brown sunken lesions on the stem and tap root most frequently near the soil line causing stunting growth or death of the plant.

The roots were usually dry unless the soil is wet. The tap root is quite brittle, showed shredding of the bark and broken very easily. Observations of the infected plants revealed the presence of sclerotia in the pith region and outer of the roots just below the bark region. (Plate-1 A)

4.3 Pathogen

The growth Pattern of the colony of *R. bataticola* appears to be somewhat cottony, fluffy and light brown to dark brown or grayish black in colour. The microscopic observations denote that mycelial cells of hyphae are septate, branched and light to dark brown in colour. Sclerotia are brown to jet black and variable in shape i.e. round to oval and irregular. (Plate-1 B, C, D, E)

Plate-1: Plant showing symptoms of dry root rot affected by *Rhizoctonia bataticola*

A. Dry root rot affected chickpea plant

➤ **Growth pattern and pigmentation of isolate of *Rhizoctonia bataticola* on Potato dextrose agar medium**

B. Front view

C. Dorsal view

➤ **Morphological characters of *Rhizoctonia bataticola* as observed in microscope**

D. Septation of mycelia

E. Formation of sclerotia



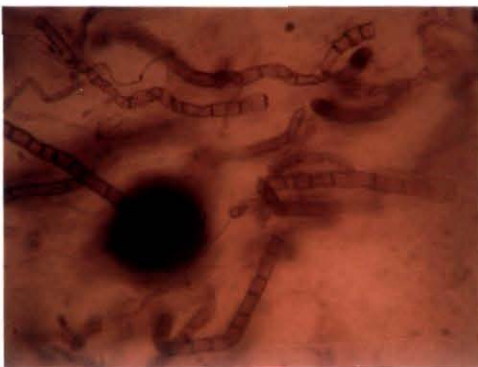
A



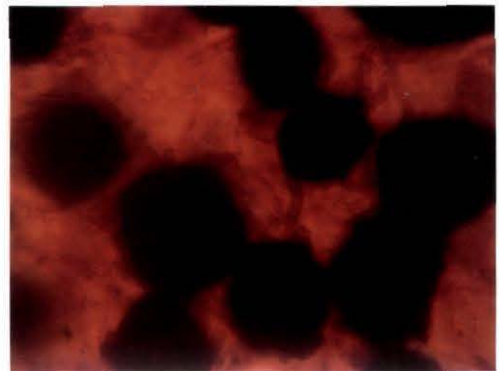
B



C



D



E

Plate 1.

4.4 Isolation, Purification and Identification of *Rhizoctonia bataticola*

Isolation from dry root rot affected plants was done on PDA medium. The Pathogen was identified on the basis of morphological characters. The culture was purified by using the hyphal tip technique and incubated at 25⁰C for further growth. The purified culture was maintained on PDA slants.

4.5 Pathogenicity test

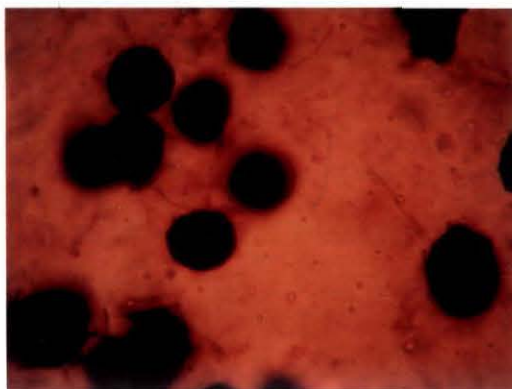
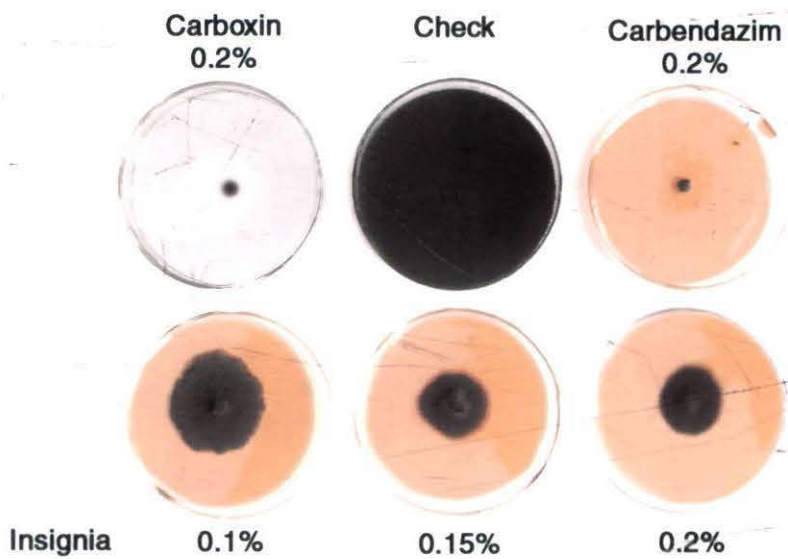
The pathogenicity test was conducted by using Blotter Paper technique (Nene *et al.* 1981). Symptoms on roots appeared after 08 days of incubation at 30°C. The roots exhibited brown to black discolourtion.

4.6 Evaluation of fungicides on growth and Sclerotia formation of *Rhizoctonia bataticola*

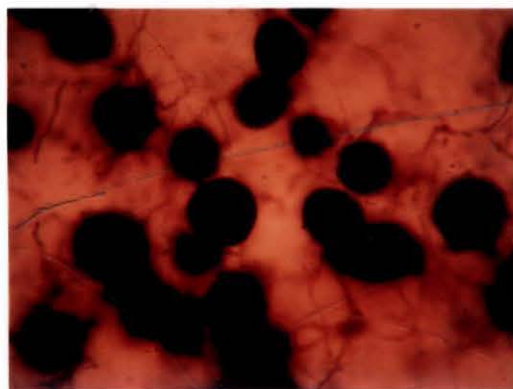
The results of *in vitro* testing of fungicides on the growth and sclerotia formation of *R. bataticola* presented in Table-1 showed that Vitavax (Carboxin) and Bavistin (Carbendazim) completely inhibited the mycelial growth and sclerotia formation at 0.2 per cent concentration. However, in all the three concentrations of Insignia i.e. 0.1, 0.15 and 0.2 per cent the colony diameter observed as 43.6, 28.6 and 25.6 mm respectively on the 5th day. Minimum sclerotial count/5 mm disc (5) was recorded at 0.2 per cent concentration of Insignia. Petriplates in the check was completely covered with mycelial growth on 5th day and sclerotia formation was also maximum (28) on 10th day. (Plate-2)

Plate-2: *In vitro* Evaluation of fungicides by poisoned food technique

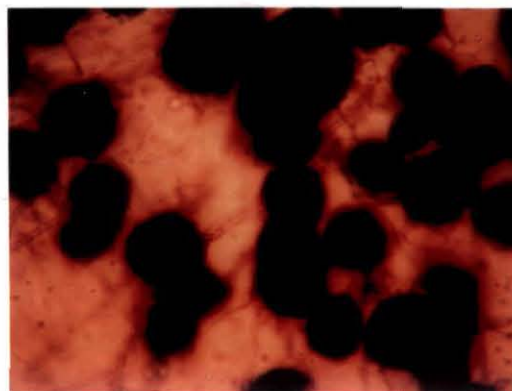
- A. Petri plates of insignia at three concentrations, Carboxin, Carbendazim and control
- B. Sclerotia formation of *R. bataticola* at three concentration of Insignia alongwith control



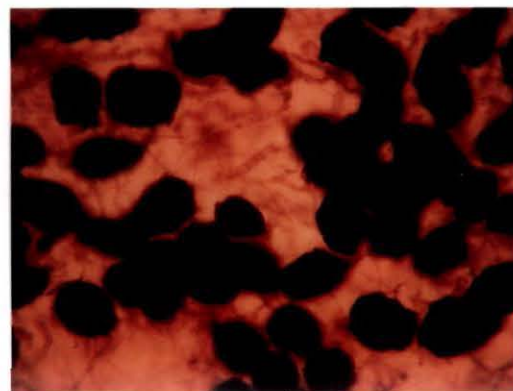
0.1 Per cent Insignia



0.15 Per cent Insignia



0.2 Per cent Insignia



Check

Table 1. Evaluation of fungicides on growth and Sclerotia formation of *Rhizoctonia bataticola*

S. No	Fungicides	Concentration(%)	* Colony diameter (mm)					**Sclerotia formation	
			1 st day	2 nd day	3 rd day	4 th day	5 th day	5 th day	10 th day
1.	Insignia (Polyclostrabin)	0.1	11.1	28.9	33.1	40.6	43.6	-	11
2.	Insignia (Polyclostrabin)	0.15	10.2	15.4	23.0	25.6	28.6	-	8
3.	Insignia (Polyclostrabin)	0.2	7.4	12.2	22.5	24.0	25.6	-	6
4.	Vitavax (Carboxin)	0.2	0.0	0.0	0.0	0.0	0.0	-	-
5.	Bavistin (Carbendazim)	0.2	0.0	0.0	0.0	0.0	0.0	-	-
6.	Check	-	20.6	47.5	66.5	78.5	90.0	-	28

* Mean of 6 replications

** Mean of 10 microscopic observations

- No sclerotia formation

4.7 Evaluation of commercial products of bioagent (*Trichoderma viride*) on growth and Sclerotia formation of *Rhizoctonia bataticola* (in vitro)

Results of *in vitro* evaluation of *Trichoderma viride* against *R. bataticola* by dual culture technique exhibited in Table-2 revealed that all the three commercial products viz; Neemoderma, Niprot and Trioseal were effective in inhibiting the growth of *R. bataticola* as compared to control. Almost they varied in growth pattern having 53, 47 and 33 mm respectively on 5th day. Growth of *R. bataticola* also distinct (37, 43 and 57 mm) in all commercial products. Neemoderma was most effective in inhibiting mycelial growth as well as sclerotia formation.

Sclerotia formation of *R. bataticola* recorded after 5 and 10 days through microscopic observation exhibited no sclerotia formation on the 5th

Plate- 3: Evaluation of commercial products of bioagent (*Trichoderma viride*) by dual culture technique

- A. Plate showing growth of *T. viride* of three commercial products and pathogen.
- B. Plates showing antagonistic effect of *T. viride* on 10th day.

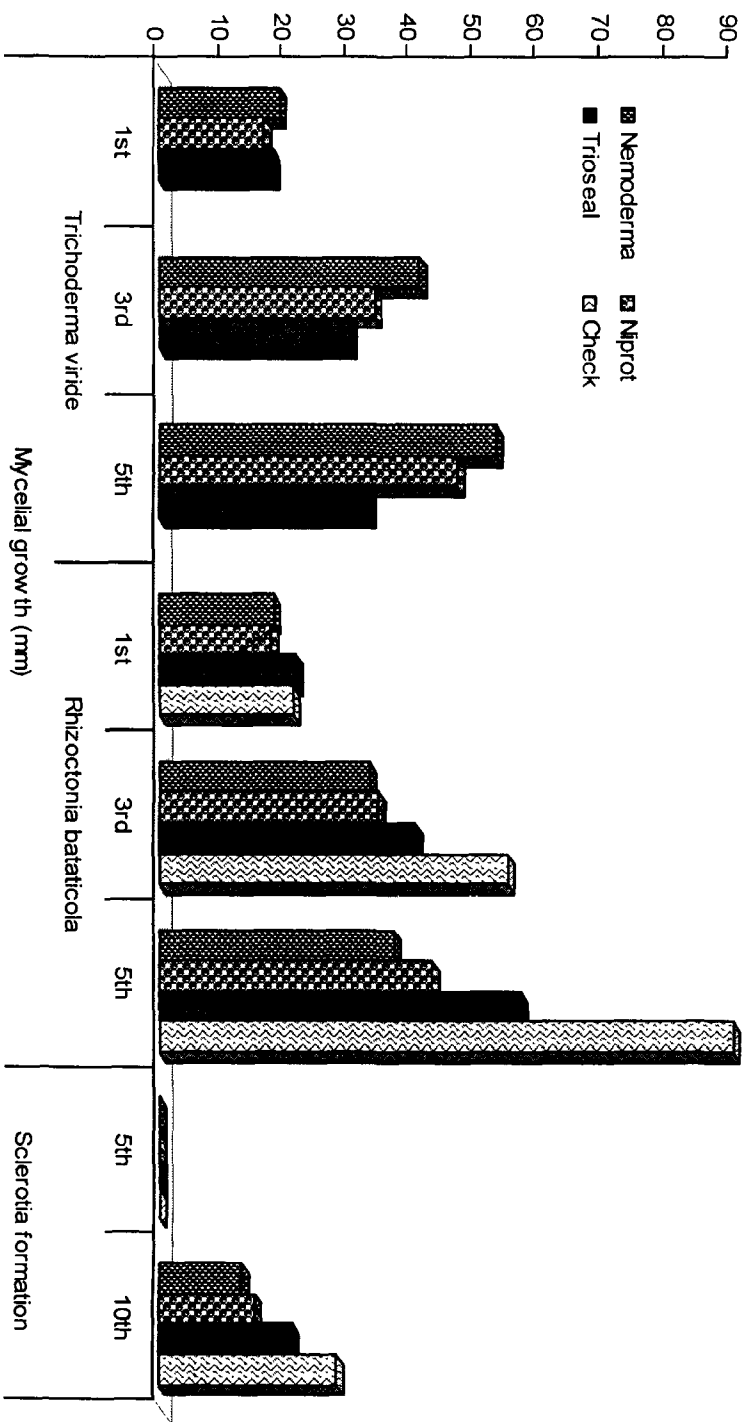


Fig. 1. Evaluation of commercial products of biocontrol agent i.e. *T. viride* on growth and sclerotia formation of *Rhizoctonia bataticola*

day. However, on 10th day it was minimum (13) in Neemoderma followed by Niprot (15) and Trio seal (21). The mycelial growth of *R. bataticola* was maximum (90mm) in the check after 5 days and maximum sclerotia formation (28) after 10 days. According to Bell's reaction given in the Table-2 Neemoderma (R₄) and Niprot (R₄) were the effective commercial products as it blocked at point of contact of *R. bataticola* but Trio seal (R₅) allowed *R. bataticola* to over grow. (Plate-3 and Fig-1)

Table 2. Evaluation of commercial products of bioagent (*Trichoderma viride*) on growth and Sclerotia formation of *Rhizoctonia bataticola* (in vitro)

S. No	Commercial Product	Mycelial growth (mm)						**Sclerotia formation		Bell's scale
		<i>*Trichoderma viride</i>			<i>*Rhizoctonia bataticola</i>					
Days		1 st	3 rd	5 th	1 st	3 rd	5 th	5 th	10 th	
1	Neemoderma	19.0	41.0	53.0	18.0	33.0	37.0	-	13	R ₄
2	Niprot	16.5	34.0	47.0	17.5	34.5	43.0	-	15	R ₄
3	Trio seal	18.0	30.0	33.0	21.5	40.5	57.0	-	21	R ₅
4	Check				21.0	55.0	90.0	-	28	

* Mean data of three replications

** Mean of 10 microscopic observations

- No sclerotia formation

Bell's reaction:

R1 = 100% over growth of *T. viride*,

R2 = 75% over growth,

R3 = 55% over growth,

R4 = Block at point of contact,

R5 = Pathogen over grow antagonist.

Plate - 4: Evaluation of chickpea entries against dry root rot by using Blotter paper technique

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3	Trio seal	18.0	30.0	33.0	21.5	40.5	57.0	-	21	R ₅
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Days		1 st	3 rd	5 th	1 st	3 rd	5 th	5 th	10 th	
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3	Trio seal	18.0	30.0	33.0	21.5	40.5	57.0	-	21	R ₅
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		<i>*Trichoderma viride</i>			<i>*Rhizoctonia bataticola</i>					
Days		1 st	3 rd	5 th	1 st	3 rd	5 th	5 th	10 th	
1	Neemoderma	19.0	41.0	53.0	18.0	33.0	37.0	-	13	R ₄
2	Niprot	16.5	34.0	47.0	17.5	34.5	43.0	-	15	R ₄
3	Trio seal	18.0	30.0	33.0	21.5	40.5	57.0	-	21	R ₅
4	Check				21.0	55.0	90.0	-	28	

* Mean data of three replications

** Mean of 10 microscopic observations

- No sclerotia formation

Bell's reaction:

R1 = 100% over growth of *T. viride*,

R2 = 75% over growth,

R3 = 55% over growth,

R4 = Block at point of contact,

R5 = Pathogen over grow antagonist.

day. However, on 10th day it was minimum (13) in Neemoderma followed by Niprot (15) and Trio seal (21). The mycelial growth of *R. bataticola* was maximum (90mm) in the check after 5 days and maximum sclerotia formation (28) after 10 days. According to Bell's reaction given in the Table-2 Neemoderma (R₄) and Niprot (R₄) were the effective commercial products as it blocked at point of contact of *R. bataticola* but Trio seal (R₅) allowed *R. bataticola* to over grow. (Plate-3 and Fig-1)

Table 2. Evaluation of commercial products of bioagent (*Trichoderma viride*) on growth and Sclerotia formation of *Rhizoctonia bataticola* (in vitro)

S. No	Commercial Product	Mycelial growth (mm)						**Sclerotia formation		Bell's scale
		<i>*Trichoderma viride</i>			<i>*Rhizoctonia bataticola</i>					
Days		1 st	3 rd	5 th	1 st	3 rd	5 th	5 th	10 th	
1	Neemoderma	19.0	41.0	53.0	18.0	33.0	37.0	-	13	R ₄
2	Niprot	16.5	34.0	47.0	17.5	34.5	43.0	-	15	R ₄
3	Trio seal	18.0	30.0	33.0	21.5	40.5	57.0	-	21	R ₅
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4	Check				21.0	55.0	90.0	-	28	

- * Mean data of three replications
- ** Mean of 10 microscopic observations
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Bell's reaction:

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* Mean data of three replications

** Mean of 10 microscopic observations

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Bell's reaction:

R₁ = 100% over growth of *T. viride* ,

R₂ = 75% over growth,

R₃ = 55% over growth,

R₄ = Block at point of contact,

R₅ = Pathogen over grow antagonist.

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Table 2. Evaluation of commercial products of bioagent (*Trichoderma viride*) on growth and Sclerotia formation of *Rhizoctonia bataticola* (in vitro)

S. No	Commercial Product	Mycelial growth (mm)						**Sclerotia formation		Bell's scale
		<i>*Trichoderma viride</i>			<i>*Rhizoctonia bataticola</i>					
Days		1 st	3 rd	5 th	1 st	3 rd	5 th	5 th	10 th	
1	Neemoderma	19.0	41.0	53.0	18.0	33.0	37.0	-	13	R ₄
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* Mean data of three replications

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R2 = 75% over growth,

R3 = 55% over growth,

R4 = Block at point of contact,

R5 = Pathogen over grow antagonist.



Growth of *R. bataticola* on PDB



Seedlings of test line inoculated with *R. bataticola* in an ET Chamber



Control: Uninoculated seedling



Susceptible check (BG-212)

4.8 Host resistance: - Screening of chickpea entries/genotypes against dry root rot

4.8.1. Blotter Paper Technique

To locate the sources of host resistance against dry root rot pathogen 143 entries/genotypes and 44 germplasm accessions were collected from AICRP on chickpea, JNKVV, Jabalpur. These were screened alongwith susceptible check BG 212 through Blotter paper technique and further summarized on the basis of per cent mortality as resistant, Moderately resistant, Susceptible and Highly susceptible. The observations recorded on 8th day are presented in Tables 3.1. to 3.9 Disease reaction of pathogen on chickpea entries/genotypes is shown in plate 4 and 5.

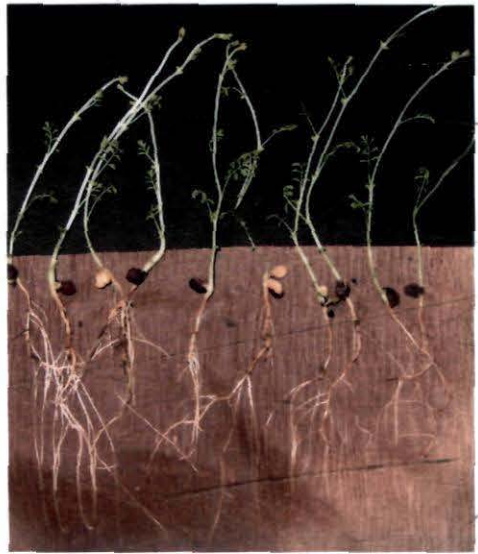
Evaluation of chickpea entries against dry root rot by using Blotter paper technique

Table 3.1 National nursery: Entries - 49

S.No	Genotype	Rating	Category	DRR (%)
1	NND-1	7	S	35
2	NND-2	7	S	30
3	NND-3	7	S	35
4	NND-4	5	T	20
5	NND-5	3	M R	15
6	NND-8	3	M R	20
7	NND-9	5	T	25
8	NND-13	5	T	20
9	NND-15	5	T	20



Resistant (R) : 1



Moderately Resistant (MR) : 3



Tolerant (T) : 5



Susceptible (S) : 7



Highly susceptible (HS) : 9

10	NND-16	5	T	25
11	NND-17	5	T	20
12	NND-23	3	M R	15
13	NND-26	3	M R	10
14	NND-27	9	HS	100
15	NND-28	9	HS	100
16	NND-29	5	T	20
17	NND-30	3	M R	15
18	NND-36	9	HS	100
19	NND-37	7	S	35
20	NND-38	7	S	40
21	NND-39	7	S	35
22	NND-40	5	T	20
23	NND-42	5	T	25
24	NND-43	3	M R	10
25	NND-44	1	R	0
26	NND-45	7	S	35
27	NND-46	9	HS	100
28	NND-47	5	T	25
29	NND-48	3	MR	10
30	NND-49	1	R	0
31	NND-51	7	S	35

32	NND-52	5	T	20
33	NND-53	9	HS	100
34	NND-54	1	R	0
35	NND-55	5	T	20
36	NND-56	1	R	0
37	NND-58	9	HS	100
38	NND-60	3	MR	15
39	NND-61	5	T	25
40	NND-62	3	MR	15
41	NND-63	3	MR	10
42	NND-64	5	T	20
43	NND-67	7	S	35
44	NND-68	3	MR	15
45	NND-70	3	MR	10
46	NND-71	5	T	20
47	NND-72	1	R	0
48	NND-73	3	MR	15
49	NND-74	5	T	25

*Average of two replications

Table 3.2 Confirmation of Dry root rot resistance: Entries -17

S.No	Genotype	Rating	Category	DRR (%)
1	P-202	5	T	25
2	P-205	3	MR	20
3	P-207	9	HS	100
4	P-210	5	T	25
5	P-211	3	MR	20
6	P-212	3	MR	10
7	P-213	3	MR	10
8	P-214	9	HS	100
9	P-216	7	MS	35
10	P-217	9	HS	100
11	P-218	3	MR	10
12	P-220	9	HS	100
13	P-223	5	T	20
14	P-224	1	R	0
15	P-226	3	MR	15
16	P-227	3	MR	10
17	P-228	1	R	0

*Average of two replications

Table3.5 IVT (Desi): Entries - 26

S.No	Genotype	Rating	Category	DRR (%)
1	P-1	3	MR	10
2	P-2	7	S	30
3	P-3	5	T	25
4	P-4	3	MR	15
5	P-5	9	HS	100
6	P-6	3	MR	15
7	P-7	3	MR	20
8	P-8	5	T	20
9	P-9	3	MR	10
10	P-10	7	S	35
11	P-11	1	R	0
12	P-12	9	HS	100
13	P-13	5	T	20
14	P-14	5	T	10
15	P-15	9	HS	100
16	P-16	3	MR	15
17	P-17	5	T	25
18	P-18	5	T	30
19	P-19	7	S	35
20	P-20	3	MR	15
21	P-21	9	HS	100
22	P-22	3	MR	20
23	P-23	1	R	0
24	P-24	5	T	25
25	P-25	3	MR	15
26	P-26	5	T	25

*Average of two replications

Table 3.3 AVT 1 (Large seeded): Entries - 04

S.No	Genotype	Rating	Category	DRR (%)
1	P-30	9	HS	100
2	P-31	9	HS	100
3	P-32	3	MR	20
4	P-33	7	S	35

*Average of two replications

Table 3.4 AVT 1 (ELSK): Entries – 06

S.No	Genotype	Rating	Category	DRR (%)
1	P-115	5	T	15
2	P-116	3	MR	5
3	P-118	5	T	10
4	P-119	1	R	0
5	P-121	9	HS	100
6	P-123	9	HS	100

*Average of two replications

Table 3.6 Promising entries: Entries – 16

S.No	Genotype	Rating	Category	DRR(%)
1	JSC-37	1	R	0
2	HKO2-207	1	R	0
3	HO2-14	5	T	20
4	JG2003-14-16	1	R	0
5	HO3-H5	3	MR	10
6	HK-03-11-3	3	MR	15
7	MPJG-2004-03	5	T	20
8	JG-1-14	5	T	20
9	JG-2-125	3	MR	5
10	JG-2001-115	1	R	0
11	MPJG-2001-04	1	R	0
12	MPJG-99-115	5	T	15
13	JG-63	1	R	0
14	MPJG-89-11551	1	R	0
15	MPJG-89-9023	3	MR	10
16	JG-1-9	3	MR	10

*Average of two replications

Table 3.7 AVT 1 (Late sown): Entries - 04

S.No	Genotype	Rating	Category	DRR(%)
1	P-41	3	MR	10
2	P-42	9	HS	100
3	P-43	5	T	25
4	P-44	5	T	20

*Average of two replications

Table 3.8 IVT (Rainfed): Entries - 21

S.No	Genotype	Rating	Category	DRR(%)
1	P-75	5	T	20
2	P-76	7	S	35
3	P-77	5	T	15
4	P-78	1	R	0
5	P-79	9	HS	100
6	P-80	3	MR	20
7	P-81	5	T	25
8	P-82	5	T	25
9	P-83	3	MR	10
10	P-84	7	S	35
11	P-85	5	T	10
12	P-86	5	T	15

13	P-87	9	HS	100
14	P-88	1	R	1
15	P-89	7	S	30
16	P-90	5	T	20
17	P-91	3	MR	10
18	P-92	5	T	25
19	P-93	7	S	35
20	P-94	5	T	20
21	P-95	5	T	25

*Average of two replications

Table 3.9 Germplasm lines: Entries - 44

S.No	Genotype	Rating	Category	DRR(%)
1	IC-251669	3	MR	10
2	IC-251706	7	S	15
3	IC-251710	7	S	25
4	IC-251713	5	T	20
5	IC-251714	5	T	10
6	IC-251726	1	R	0
7	IC-251733	1	R	0
8	IC-251740	3	MR	15
9	IC-251741	1	R	0

10	IC-251754	3	MR	10
11	IC-261813	3	MR	10
12	IC-251889	7	S	30
13	IC-255449	3	MR	15
14	IC-255466	1	R	0
15	IC-327290	5	T	20
16	IC-327204	5	T	15
17	IC-327446	7	S	25
18	IC-327083	5	T	15
19	IC-327933	9	HS	100
20	ICC-03104	5	T	20
21	ICC-03112	3	MR	10
22	ICC-04107	1	R	0
23	ICC-11322	5	T	20
24	ICC-12467	3	MR	15
25	ICC-14374	1	R	0
26	ICC-14367	5	T	15
27	ICC -14433	3	MR	10
28	ICC-14391	1	R	0
29	ICC-269359	1	R	0
30	IC-269511	5	T	15
31	IC-269514	5	T	20

32	IC-269614	7	S	30
33	IC-269671	7	S	35
34	IC-269685	5	T	15
35	IC-269737	3	MR	10
36	IC-269750	1	R	0
37	IC-269785	3	MR	10
38	IC-269826	5	T	20
39	IC-269842	9	HS	100
40	ICCV-97125	3	MR	10
41	ICCV-04109	7	S	35
42	ICCV-06107	7	S	30
43	ICC-06103	5	T	20
44	ICC-05106	7	S	35

*Average of two replications

The entries/genotype has been graded on the basis of rating scale- (1-9)

Table 4. Performance of entries/genotype against dry root rot

Dry root rot Percent (%)	Entries	Grading	No. of entries
0.0 to 10	NND-44, NND-49, NND-54, NND-56, NND-72, P209, P224, P228, P119, P11, P23, JSC-37, JG2004-14-16, P-78, P-88, IC-251726, IC-251733, IC-251741, IC-255466, JG-2001-115, MPJG-2001-04, JG-63, ICC-0410, ICC-14374, ICC-14391, MPJG-89-11551, ICC-269359, IC-269750.	Resistant	28
10.1 to 20	NND-5, NND-8, NND-23, NND-26, NND-30, NND-43, NND-48, NND-60, NND-62, NND-63, NND-68, NND-70, NND-73, P205, P212, P213, P218, P226, P227, P32, P116, P1, P4, P6, P9, P16, P20, P22, P25, H03-H5, HK-03-11-3, P41, P80, P91, IC-251669, IC-251740, IC-251754, IC-261813, IC-261813, IC-255449, JG-1-9, JG-2-125, ICC-03112, ICC-12467, MPJG-89-9023, IC-269737, IC-269785, ICCV-97125.	Moderately Resistant	53
20.1 to 30	NND-4, NND-9, NND-13, NND-15, NND-16, NND-17, NND-29, NND-40, NND-42, NND-47, NND-52, NND-59, NND-61, NND-64, NND-71, NND-74, P223, P202, P210, P223, P115, P118, P3, P8, P13, P14, P17, P18, P24, HO2-14, P43, P44, P75, P77, P81, P82, P85, P86, P90, P92, P94, P95, IC-251713, IC-251714, IC-327290, IC-327204, IC-327083, MPJG-2004-03, JG-1-14, MPJG-99-115, ICC-03104, ICC-11322, ICC-14367,	Tolerant	51

	IC-269511, IC-269514, IC-269685, IC-269826, ICC-06103.		
30.1 to 40	NND-1, NND-2, NND-3, NND-37, NDD-38, NND-39, NND-45 NND-51, NND-67, NND-217, P33, P2, P10, P19, P76, P84, P89, P93, IC-251706, IC-251710, IC-251889, IC-327446 IC-269614, IC-269671, ICCV-04109, ICCV-06107 ICC-05106	Susceptible	33
Above to 40	NND-27, NND-28, NND-36, NND-46, NND-53, NND-58, P207, P216, P214, P217, P220, P30, P31, P121, P123, P5, P12, P15, P21, P42, P79, P87, IC-327933,	Highly Susceptible	23
	Total		187

4.8.2. Disease sick field

To locate the sources of host resistance against dry root rot entries/genotypes were screened alongwith susceptible BG 212 and resistance JG315 checks under the disease sick field in two replications. Observations in per cent mortality due to disease recorded from flowering to pod filling stage and the data is presented in (table -13 and plate -4).

The disease was observed in the range of 3.5 (P201) to 60.0 (MPJG99-115) per cent. The data exhibits that among all the entries, 05 entries namely JG-74, JG315, KWR-108, P-201, JG-63 showed less than 10 per cent mortality and may be considered as resistant, 14 entries namely IPC97-29, DCP92-3, Annegiri, IPC20004-52, C-104, K-850, P-205, P-209, P-221, P-222, P-223, JG-2-125, JG2001-115 exhibited 10.0 to 20.0 per cent mortality and graded as moderately resistant and 07 entries namely Chaffa, GPF-2, P-207, P-211, MPJG2004-03, JG-1-14, MPJG2001-04, exhibited 20 to 30 per cent and comes under tolerant group rest of the entries susceptible and highly susceptible.

Plate - 6: Evaluation of chickpea entries against dry root rot in disease sick field

	IC-269511, IC-269514, IC-269685, IC-269826, ICC-06103.		
30.1 to 40	NND-1, NND-2, NND-3, NND-37, NDD-38, NND-39, NND-45 NND-51, NND-67, NND-217, P33, P2, P10, P19, P76, P84, P89, P93, IC-251706, IC-251710, IC-251889, IC-327446 IC-269614, IC-269671, ICCV-04109, ICCV-06107 ICC-05106	Susceptible	33
Above to 40	NND-27, NND-28, NND-36, NND-46, NND-53, NND-58, P207, P216, P214, P217, P220, P30, P31, P121, P123, P5, P12, P15, P21, P42, P79, P87, IC-327933,	Highly Susceptible	23
	Total		187

4.8.2. Disease sick field

To locate the sources of host resistance against dry root rot entries/genotypes were screened alongwith susceptible BG 212 and resistance JG315 checks under the disease sick field in two replications. Observations in per cent mortality due to disease recorded from flowering to pod filling stage and the data is presented in (table -13 and plate -4).

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Plate 6.

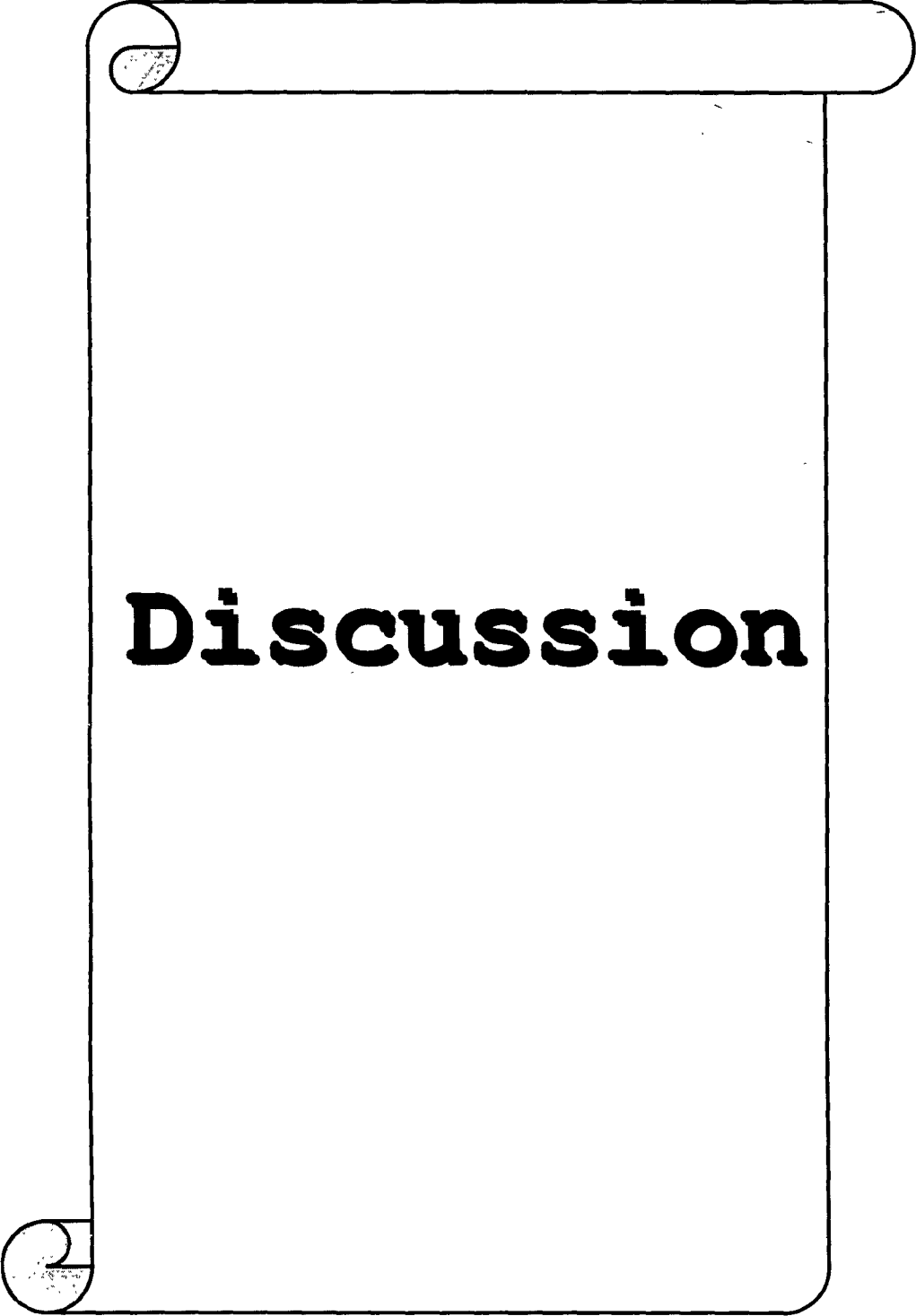
**Table 5. Evaluation of chickpea entries against dry root rot under
disease sick field**

S. No	Entries	Per cent mortality		
		R1	R2	Average
1	IPC97-29	13.9	24.0	18.9
2	DCP92-3	12.0	09.0	10.5
3	Annegiri	12.0	16.0	14.0
4	JG74	00.0	20.0	10.0
5	IPC2004-52	10.0	24.0	17.0
6	CPS-1	26.0	40.0	33.0
7	Chaffa	22.2	23.8	23.0
8	C-104	11.4	16.0	13.7
9	K-850	06.6	18.7	12.6
10	JG-62	-	-	-
11	JG-315	03.0	14.0	08.5
12	JG-63	20.0	27.0	23.5
13	KWR108	07.6	12.0	09.8
14	GPF2	27.0	20.0	23.5
15	JG1-9	42.3	26.6	34.4
16	MPJG2004-03	23.6	18.1	20.8
17	JG1-14	38.8	10.1	24.8

18	JG2-125	05.2	32.5	18.8
19	JG2001-115	15.2	14.0	14.6
20	MPJG2001-04	26.6	14.5	20.5
21	MPJG99-115	11.1	23.9	60.0
22	P201	04.0	03.0	03.5
23	P204	22.0	42.6	32.3
24	P205	22.0	14.8	18.4
25	P207	17.5	30.1	23.8
26	P206	28.5	22.4	52.4
27	P209	10.7	27.7	19.2
28	P211	31.4	27.5	29.4
29	P215	37.1	42.5	39.8
30	P216	28.6	31.6	30.2
31	P217	35.7	34.0	34.8
32	P221	10.8	16.9	13.8
33	P222	10.5	14.2	12.3
34	P223	14.4	15.6	15.0

Table 6. The entries / varieties can be graded on the basis of percent mortality due to dry root rot

Dry root rot Percent (%)	Entries	Grading	No. of entries
0.0 to 10	JG-74, JG315, KWR-108, P-201, JG-63.	Resistant	05
10.1 to 20	IPC97-29, DCP92-3, Annegiri, IPC20004-52, C-104, K-850, P-205, P-209, P-221, P-222, P-223, JG-2-125, JG2001-115.	Moderately Resistant	14
20.1 to 30	Chaffa, GPF-2, P-207, P-211, MPJG2004-03, JG-1-14, MPJG2001-04.	Tolerant	07
30.1 to 40	CPS-1, P-204, P-215, P-216, P-217, JG-1-9.	Susceptible	06
Above to 40	P-206, MPJG99-115	Highly Susceptible	02
	Total		34



Discussion

DISCUSSION

Dry root rot of chickpea caused by *Rhizoctonia bataticola* is the most important disease and causes enormous losses at podding and maturity stage when the temperature exceeds $>30^{\circ}\text{C}$. It is soil borne pathogen and can be managed by integration of the management practices including seed treatment with fungicides, bioagents, host resistance and cultural practices.

Various market available products of bioagents (*Trichoderma viride*) are being widely used by the farmers alone or in combination of fungicides without knowing their efficacy. Hence, investigations were undertaken with a view to know the most effective products of *Trichoderma viride* to be used as seed treatment or soil application and to locate the host resistance to minimize the losses from dry root rot.

The disease appeared in post flowering and pod formation stage and ranged from 3.5 to 60 per cent. Nema and Khare (1973) reported that chickpea cultivation in Madhya Pradesh was greatly threatened by soil borne diseases. Gupta *et al.* (1983) reported the incidence of root rot ranging from 3.2 to 20.6 per cent in 30 villages of northern Madhya Pradesh. Nene *et al.* (1984) also recorded heavy losses to chickpea crop due to dry root rot caused by *R. bataticola*.

Dry root rot symptoms observed in diseased plants exhibited straw colour appearance and tap root mainly becomes brittle having minute sclerotia. These symptoms separated this disease from the so-called "Wilt complex" as stated by various workers in India. (Nene *et al.* 1989). Maximum atmospheric temperature up to 34.5°C favoured the disease development. Substantial mortality and losses in chickpea due to *R. bataticola* had been

reported in high ambient temperature 30°C and above in the post – flowering stage (Singh and Mehrotra 1982).

With the view to find out effective seed dressing fungicides, three fungicides were tested *in vitro* for their effectiveness against *R. bataticola*. The results indicated that Vitavax (Carboxin) and Bavistin (Carbendazim), @ 0.2 per cent completely inhibited the growth and sclerotia formation of *R. bataticola*. However, Insignia (Polyclostrabin) @ 0.1, 0.15 and 0.2 per cent resulted partial inhibition of the mycelial growth and sclerotia formation. Higher dose of this fungicides was effective in inhibiting the mycelial growth. Taneja and Grove (1982) tested four fungicides against *R. bataticola* *in vitro* and found Benomyl and Bavistin to be best. Seed treatment with Bavistin and Vitavax (0.2%) significantly reduce the incidence of dry root rot of chickpea. These results are in agreement with the finding of Sindhan *et al* (1998) and Lambhate *et al.* (2002) reported Carbendazim, Carboxin and Topsin-n (Thiophonat-methylal) completely inhibited the growth of *R. batatiola*.

In order to make recommendation to growers and to find out most effective commercial product of *T. viride* three products were tested for their antagonistic activity against *R. bataticola* by dual culture technique. The results revealed that the comparative efficacy of all the three commercial products viz, Neemoderma, Niprot and Trioseal were almost different in growth pattern i.e. 53, 47 and 33 mm respectively on 5th day.

Neemoderma was most effective in inhibiting of mycelial growth and sclerotia formation of *R. bataticola* and causing lysis at the point of contact. Selvarajan and jeyarajan (1996) reported that *T. viride*, *T.harazianum* and *B. subtilis* formed inhibition zones against chickpea root rot pathogens *F. solani* and *R. bataticola*. Suryawanshi and Mandhare

(2009) stated *R. bataticola* was inhibited to the extent of 76% by *T. viride* and *T. harzianum*, Saxena and Saxena (1995) also reported the effectiveness of *T. viride* and *T. harzianum* against root rot of chickpea caused by *M. phaseolina*.

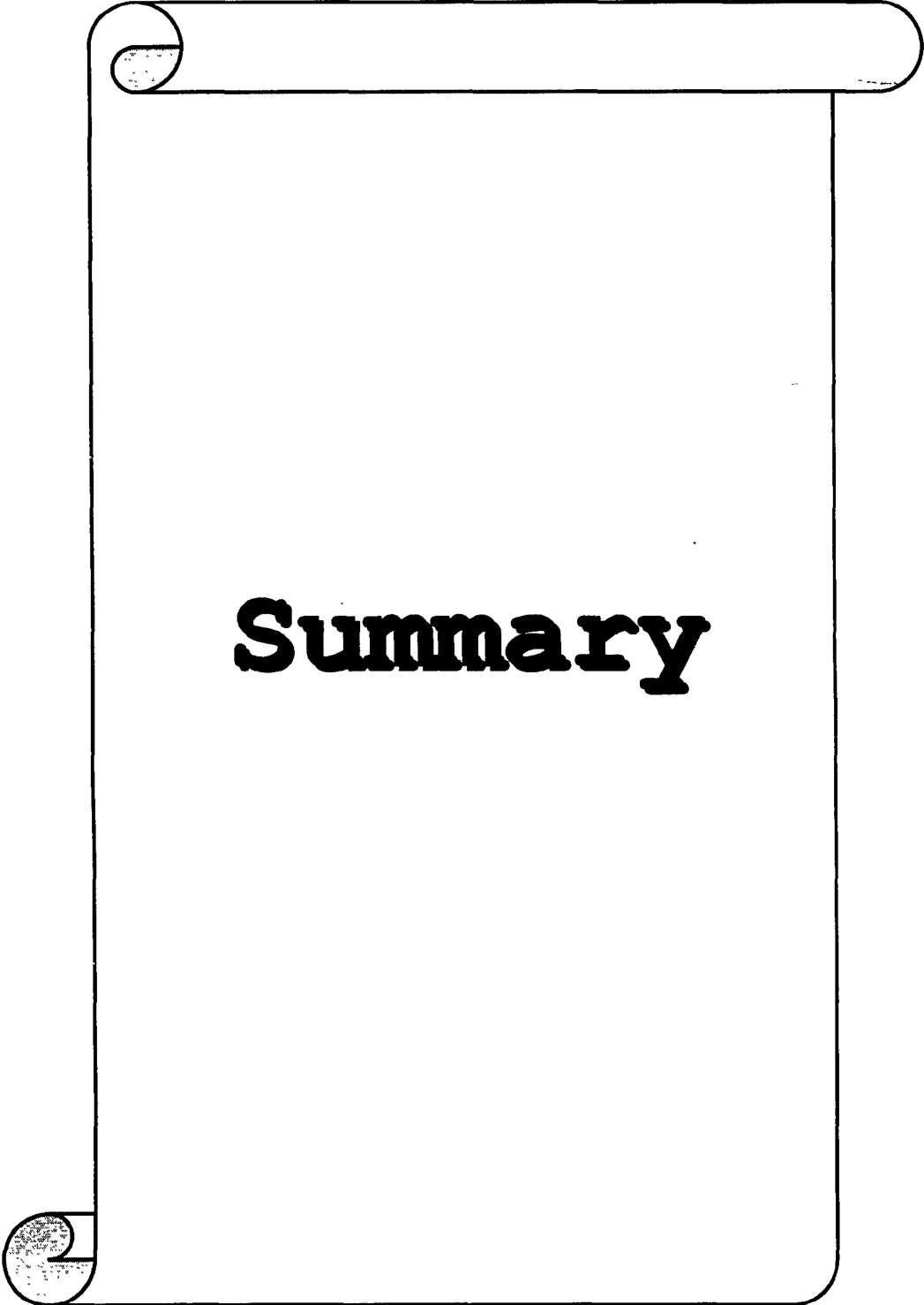
Chickpea entries/genotypes alongwith other promising material and germplasm lines were evaluated through Blotter paper technique as described by Nene *et al* (1981). In all, of the 187 entries/genotypes evaluated for host resistance have been graded as resistant, moderately resistant, tolerant, moderately susceptible and susceptible (1-9 point rating scale). Thirty entries are being scored as resistant against dry root rot. There was no blackening of the root system as it was quite healthy, completely developed and exhibited white appearance and showed <10 per cent mortality.

Furthermore, in 38 entries the per cent mortality due to root rot has been observed in the range 10-20 per cent and graded as moderately resistant. Chickpea genotypes have been screened and identified as resistant to dry root rot by various workers on the basis of *in vitro* screening (Nene *et al.*, 1981, Anand Rao and Haware, 1987; Khalid and Ilyas, 2000; Gupta *et al.*, 2003).

Lesions on roots were seen in moderately susceptible genotypes though the new roots were generally free from these lesions. Whereas infected roots were completely discolored in highly susceptible genotype BG 212 (check). Pandey *et al.* (2004) evaluated 29 chickpea germplasm accessions for resistance against dry root rot caused by *R. bataticola* under *in vitro* conditions. He observed that 22 lines were either susceptible or highly susceptible for dry root rot.

Under field 34 entries/genotypes were screened in the artificially infested diseases sick plot, where the disease incidence ranged from 3.5 to 60 per cent. Five entries namely JG-74, JG315, KWR-108, P-201, JG-63 showed less than 10 per cent mortality and graded as resistant and 13 entries namely IPC97-29, DCP92-3, Annegiri, IPC20004-52, C-104, K-850, P-205, P-209, P-221, P-222, P-223, JG-2-125, JG2001-115 exhibited 10.0 to 20.0 per cent mortality and graded as moderately resistant.

A large number of cultivars of chickpea have been screened and identified resistant lines by various workers (Nene *et al.* 1989; Reddy *et al.* 1990; Bhatt 1993; Gupta, 1997; Yadava *et al.* 2000; Gangwar *et al.* 2002; Pandey *et al.* 2004; Gupta and Babbar, 2006;).



Summary

SUMMARY, CONCLUSIONS AND SUGGESTIONS FOR FURTHER WORK

Dry root rot of chickpea caused by *Rhizoctonia bataticola* is an important disease which varies depending upon the inoculum, physical condition of the soil and weather condition. The disease has been observed at flowering/podding stage of the crop. Diseased plants exhibited straw colour appearance with blackened roots which later on becomes brittle. Isolations from the affected plants yielded the pathogen *Rhizoctonia bataticola*.

In order to know effective fungicide and the market available products of *Trichoderma viride* for seed treatment; an *in vitro* studies revealed that among fungicides, Carboxin and Carbendazim @ 0.2 per cent completely inhibited the mycelial growth and sclerotia formation. While, polyclostrobin was less effective in inhibiting the growth. Among the three commercial products of *T. viride* evaluated against *R. bataticola* through dual culture technique. Neemoderma was found most effective in inhibiting the growth and causes lyses of the mycelium at contact.

The Blotter paper technique was employed for screening the different entries/genotypes against dry root rot. The observations recorded on disease incidence indicated that out of 187 entries/genotypes evaluated for resistance to dry root rot 30 were found to be resistant. Screening on disease sick field led to conclusion that among 34 entries, 5 were found to be resistant to dry root rot where the disease was observed in the range 3.5 to 60.0 per cent.

Conclusion

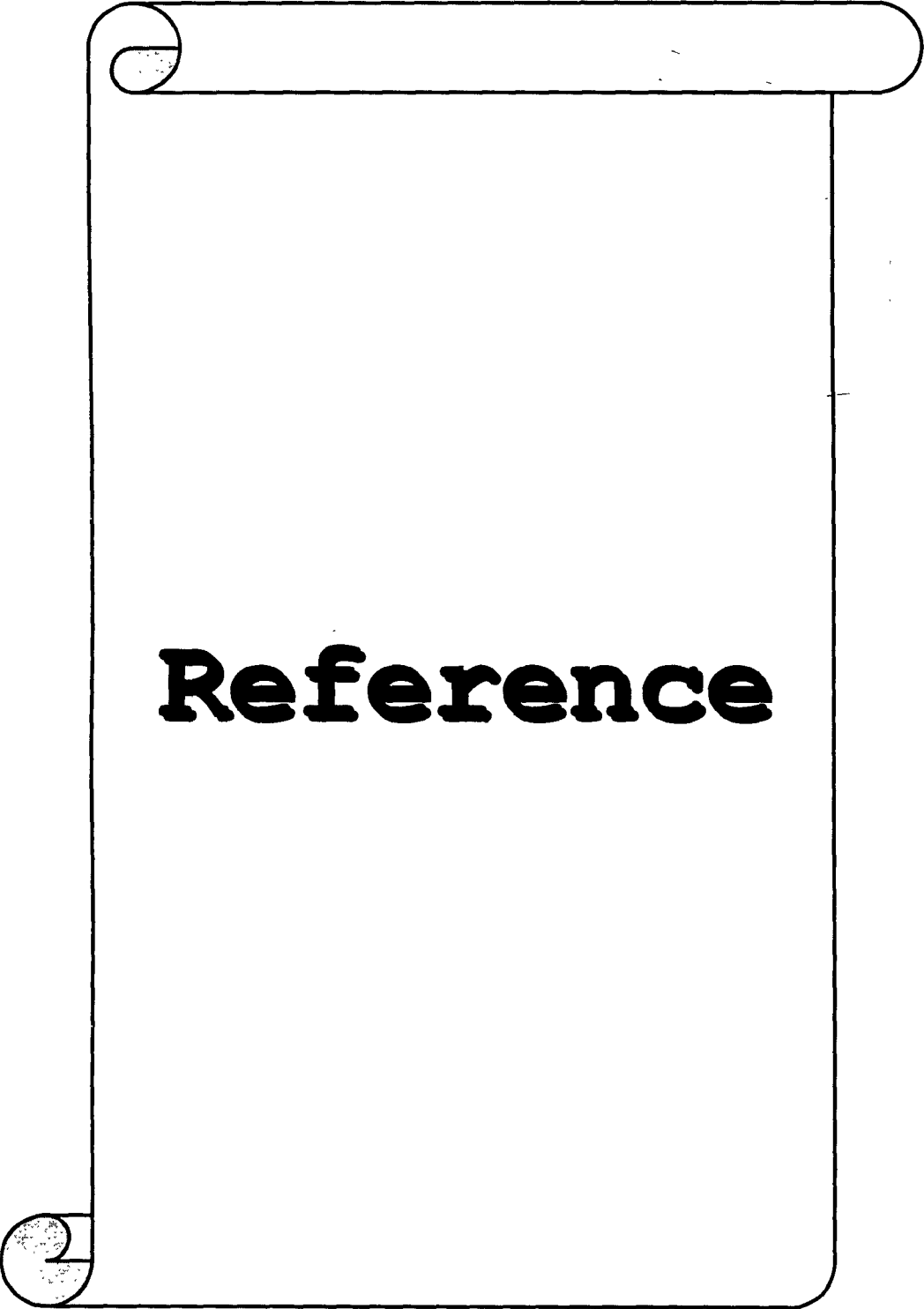
Dry root rot of chickpea caused by *Rhizoctonia bataticola* is becoming a serious problem when the ambient day temperature is exceeding $> 30^{\circ}\text{C}$ during the flowering/pod formation stage. Studies led to conclusion that fungicides found superior may be used as seed treatment in combination of bioagent (Neemoderma) effectively as they inhibited the mycelial growth and sclerotia formation *in vitro*.

In host resistance, the entries exhibited resistance under *in vitro* and disease sick field may be tested further for their confirmation against dry root rot resistance. The present study explore the possibility of using the genotype as donors parent in resistance breeding programme to incorporate the resistance to dry root rot in high yielding and wilt resistant varieties of chickpea.

Suggestions for further work

The finding made in the present investigations need to be confirmed by taking more number of commercial products of bioagents to see their comparative effectiveness. It is further suggested that local strains of bioagents should be obtained and tried alongwith commercially available products.

The entries/genotypes which were found resistant against dry root rot (*Rhizoctonia bataticola*) under *in vitro* and field conditions need to be tested in natural field conditions in infested soil under multilocation evaluation test.



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APPENDICES

Appendix:-1 Fungicides tested during investigation

Common name	Tread name	Recommended dose	Chemical group	Company
Insignia	Polyclotrabin	1, 15 & 2 g/kg	-	BASF
Carboxin	Vitabax 75% W P	2 g/kg	Oxathiins	Bayer (India) Ltd.
Carbendazim	Bavistin 50 % W P	2 g/kg	Benzimeda zole	Bayer (India) Ltd.

Appendix:-2 Details of Commercial Products of *T. viride*

S. NO.	Name of commercial products	Manufactured	Address
1	Neemoderma	Karnataka Agro-Chemicals,	180, 1 st Main Road Mahalakshmi, Layout Extension, Bangalore, India
2	Niprot	Pest Control (India) Pvt. Ltd.	Bio-control Research Laboratories (BCRL) 36/2 Sriramahalli, Nr. Rajsunkunte, Dodbollopur Road, Arakere Post, Bangalore- 560203, Karnataka, India
3	Trioseal	Central Biotech	173, Main Road, Saoner, Distt. Nagpur (MS) India

**Table 3.1 Meteorological data during the cropping season
(October 2008 – April 2009)**

Month	Week No.	Temperature (°C)		Rain Fall(mm)	Sunshine Hours	Relative Humidity	
		Mean Min.	Mean Max.			Min.	Max.
October	40	20.6	33.2	13.4	7.5	56	86
	41	18.2	34.2	00	7.8	44	83
	42	14.1	32.3	00	8.8	32	88
	43	11.1	31.0	00	9.4	26	89
November	44	10.8	32.0	00	9.0	24	85
	45	10.5	30.9	00	7.9	31	89
	46	13.2	29.6	00	4.4	37	85
	47	10.2	28.0	00	6.6	35	87
	48	07.7	27.4	00	7.3	32	90
December	49	07.4	27.7	00	7.6	31	91
	50	09.4	29.1	00	6.8	37	92
	51	09.1	27.5	00	6.3	40	90
	52	06.9	25.8	00	8.0	32	90
January	01	09.6	25.9	00	6.1	45	89
	02	13.6	22.4	37.8	5.4	71	96
	03	10.6	26.8	00	8.6	42	92
	04	09.2	27.5	00	8.6	39	92
February	05	09.1	26.7	00	9.6	35	90
	06	10.1	29.9	00	9.4	33	87
	07	09.7	28.6	00	9.6	29	88
	08	11.3	31.0	00	9.4	21	78
March	09	12.3	33.1	00	9.6	21	79
	10	14.0	33.9	2.0	7.5	25	69
	11	12.6	33.4	00	6.8	18	72
	12	15.2	35.4	00	8.0	18	66
April	13	15.3	36.4	00	8.9	15	67
	14	17.3	37.7	00	9.1	21	63
	15	15.4	36.3	00	9.7	15	59
	16	20.2	41.1	00	10.0	10	43

Vita

The author of this thesis Mr. Shashank Ranjan was born on 22nd Jan., 1983 at District: Chhatarpur, (M.P.). He passed his high school (10th) in year 1999 from Sarsawati Higher Secondary School Chhatarpur, (M.P.) with Second class. He passed his higher secondary in the year 2001 from Bundelkhand higher Secondary School Chhatarpur (M.P.) with first class. He passed B.Sc. (Ag.) in year 2007 from College of Agriculture, J.N.K.V.V. (M.P.) with 68.3 per cent marks.

Then the author completed his post-graduation in 2009 from College of Agriculture, J.N.K.V.V., Jabalpur in the department of Plant pathology, with first class and this thesis is being submitted in partial fulfillment of the degree of M.Sc. (Ag.)